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How games can support misinformation education: A sociocultural perspective

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Abstract

This study uses a sociocultural perspective, which views literacy as embedded in people's daily practices and shaped by social contexts, to explore how a misinformation escape room can support learning about misinformation. While the sociocultural perspective has a rich theoretical foundation, it has rarely been used to examine, much less evaluate, information and media literacy interventions. In this paper, we posit that the topic of misinformation makes a strong case for using the sociocultural model and explore a misinformation escape room through this lens. We present findings of a nationwide study of an online misinformation escape room with post-game debrief discussion conducted at 10 public libraries that hosted 53 game sessions involving 211 players. The mixed methods study finds the game and accompanying debrief supported players in reflecting upon social media platform infrastructures, the psychological and emotional dimensions of misinformation, and how their personal behaviors intersect with online misinformation. We discuss how the sociocultural perspective can enrich our understanding of the role played by certain attributes of the game—narrative, debrief, and collaboration—thereby providing insights for the design of media and information literacy interventions.

1 | INTRODUCTION

The challenge of mitigating the spread and impact of misinformation has attracted growing attention including from the field of media and information literacy (MIL) (Caulfield, 2016; Haider & Sundin, 2022; Sullivan, 2018; Tripodi et al., 2023; Wedlake & Bugre, 2023). Numerous educational curricula, activities, checklists, and other resources have been developed to support people to navigate an increasingly fraught information environment (Caulfield, 2016; Sullivan, 2018; Young et al., 2020). The growing sophistication of technologies layered on top of a distributed, social, online ecosystem, has presented the MIL field with a formidable challenge (Haider & Sundin, 2022). How can we support people in different circumstances, backgrounds, and interests to be better prepared to traverse today's information ecosystem?

Misinformation games have emerged as one promising approach. Three recent review articles identified 44 misinformation games and more under development (Contreras-Espinosa & Eguia-Gomez, 2023; DeJong, 2023a; Glas et al., 2023). Various studies have shown the many ways games can create an engaging and low-stakes way for people to learn about misinformation (Basol et al., 2020; Roozenbeek & van der Linden, 2019; Roozenbeek & van der Linden, 2020; Wedlake & Bugre, 2023). This research examines one such game, *The Euphorigen Investigation*, a misinformation escape room game developed by co-authors Chris Coward and Jin Ha Lee, and the co-founder of the company Puzzle Break, Lindsey Morse.

This paper follows an earlier, exploratory study that showed the potential of the game and accompanying debrief to increase awareness of certain misinformation tactics and generate reflection on the psychological

dimensions of misinformation (Cho et al., 2023). In reflecting on the earlier study, we concluded that there are limitations to using measures that are commonly found in the misinformation MIL literature, such as an individual's capacity to detect misinformation. The aim of this paper is to explore how games, such as *The Euphorigen Investigation*, can support misinformation education when viewed through the sociocultural perspective, emphasizing the everyday practice and cultural context of literacy as opposed to a set of cognitive skills (Haider & Sundin, 2022). We use data from a nationwide study of the escape room to discuss the opportunities and challenges of using a sociocultural lens to assess the impact of the game. The research questions we aim to answer are: (1) How does *The Euphorigen Investigation* support learning about misinformation? and (2) What considerations can be derived from the sociocultural perspective for informing the design and study of misinformation MIL interventions?

2 | LITERATURE REVIEW

Among misinformation scholars, defining misinformation is a subject of debate (Altay et al., 2023; Ang et al., 2020; Wardle, 2023). Some scholars use misinformation as an umbrella term to include all false or misleading information (Altay et al., 2023; Lee & Jones-Jang, 2022; Roozenbeek et al., 2023; Van Bavel et al., 2021; Wardle, 2023), while others make a distinction between misinformation and disinformation, the latter to signify intent (Altay et al., 2023; Ang et al., 2020; Kapantai et al., 2021; Zannettou et al., 2019). In this research, we use misinformation as an umbrella term (Cho et al., 2023; Young et al., 2020). In our game, users have no way of discerning intent, thus the distinction is not relevant for the purposes of this intervention (Wardle, 2023). For studies about actors who spread mis/disinformation, or for developing technical interventions, the mis/disinformation distinction would be needed (Kapantai et al., 2021; Zannettou et al., 2019).

2.1 | Autonomous and sociocultural approaches to MIL

In their book, *Paradoxes of Media and Information Literacy: The Crisis of Information* (2022), Jutta Haider and Olof Sundin situate MIL within two contrasting theoretical and epistemological notions that have their roots in Brian Street's seminal work on literacy (Street, 1984). Broadly, Street divides literacy into two schools of thought. One school, the autonomous perspective,

"regards literacy as individual cognitive skills, independent of social circumstances" (Haider & Sundin, 2022, p. 53). This conceptualization of media literacy as a set of skills is "roughly aligned with behaviorist conceptions of learning as instrumental, objective, and measurable" (Manca et al., 2021, p. 3). Numerous critiques have been raised about the autonomous model for its lack of consideration of social practices or social circumstances (Haider & Sundin, 2022; Manca et al., 2021). It is also normative, often involving a specific set of criteria used to classify someone as information literate or not (Haider & Sundin, 2022).

The other school, the sociocultural perspective, views literacy "as embedded in the practices of everyday life" and in the "various cultural practices and processes of meaning-making" (Haider & Sundin, 2022, p. 53). The attention to infrastructural meaning-making is essential in that it "enables a power-sensitive, material-discursive understanding of how people, sources, and content converge and are shaped in practices that are part of infrastructural arrangements" (Haider & Sundin, 2022, p. 65). The context within which people experience information is consequential. In contrast to the autonomous perspective where the "'space' in which people gain these skills is less important than the skills themselves," the sociocultural perspective centers the "material, epistemological practices that lead to the development of literacy practices" (Manca et al., 2021, p. 3). Furthermore, this approach views literacy as situated in social practices, where "different practices imply different literacies" (Haider & Sundin, 2022, p. 58). Different communities, for instance, have different norms around the use of social media (Jenkins et al., 2006). While people still require technical and digital skills, they also need those skills to be situated within the larger sociocultural context (Jenkins et al., 2006).

2.2 | MIL and misinformation

The nature of misinformation makes a strong case for exploring MIL educational approaches through the sociocultural lens. As shown above, the sociocultural perspective requires attention to the infrastructural meaning-making. In the misinformation space, this would include awareness and understanding of the underlying technologies and tactics, the role of algorithms, and the affordances of social media platforms, all of which contribute to boosting the believability and spread of misinformation. It further requires attention to context, a notion that is widely covered in misinformation literature. Comparative studies have illustrated the extent to which national contexts influence susceptibility to misinformation and the

different ways misinformation manifests across diverse countries (Humprecht et al., 2023).

The literature on media and information literacy (and information literacy and digital literacy) and misinformation is vast, and a systematic review of the field is beyond the scope of this paper. For the purposes here, we focus on characteristics that animate the distinctions between the autonomous and sociocultural perspectives of MIL: skills, context, and the psychological dimensions of misinformation. MIL interventions are predominantly skills-based with an emphasis on the “rational choices individuals should make when they select and evaluate information” (Haider & Sundin, 2022, p. 27). Skills-based assessments use measurements such as veracity, credibility or plausibility of true/false information, ability to detect manipulation tactics, and sharing intention, among others (Blair et al., 2024; Kozyreva et al., 2024; Sirlin et al., 2021). However, interventions focusing solely on improving the discernment skills of particular individuals miss out on the broader contexts and social processes at play (Kozyreva et al., 2024; Ma, 2021). Some scholars have advocated for more holistic approaches to MIL that include teaching how internet platforms work, including financial structures, and internet cultures and ideologies (Caulfield, 2016; Wardle, 2023).

There is growing recognition of the need to address the emotive, social, and psychological dimension of misinformation in educational interventions (Sullivan, 2018; Young et al., 2020). The psychological sciences have made numerous contributions to misinformation research, showing how affordances of social media platforms and psychological tendencies can drive belief in certain information. Short, easy to understand messages, content compatible with one's worldview, and repeated exposure to posts that are liked and shared by friends all make the information “feel” true (Schwarz & Jalbert, 2021). People also tend to have confidence in their own abilities to recognize online misinformation (Barthel et al., 2016; Mitchell et al., 2019) while believing that misinformation has greater effects on other people, otherwise known as the third person effect (Jang & Kim, 2018; Wagner & Boczkowski, 2019).

One interesting dimension is the distinction, often a blurry one, between critical thinking and skepticism on the one hand, and cynicism on the other. Several studies have found that engaging in deliberative reasoning correlates with a greater ability to distinguish between real and fake news headlines (Bago et al., 2020). As summed up by Pennycook and Rand, “people fall for fake news because they fail to think; not because they think in a motivated or identity-protective way” (Pennycook & Rand, 2019, p. 48). On the other hand, skeptics critically evaluate information sources, and skepticism encourages

engagement and information seeking (Pinkleton et al., 2012). However, while skepticism encourages critical engagement with information, cynicism, a “lack of confidence in, and a feeling of distrust” (Pinkleton et al., 2012, p. 25), encourages disengagement. To increase resilience to misinformation, skepticism should be encouraged without leading to cynicism, a blurry line that will differ individual to individual and thus something that requires careful thought when designing misinformation interventions (Clayton et al., 2020; Modirrousta-Galian & Higham, 2023).

2.3 | Games-based learning

As sites of play but also learning (Gee, 2007), games have the potential to be fruitful grounds for learning about misinformation as a form of MIL. A benefit to using play as a way to learn is that play “lowers the emotional stakes of failing” and players “take risks and learn through trial and error” (Jenkins et al., 2006, p. 23). Games appeal to many types of people and can cover a wide variety of topics—including those not typically covered in formal educational settings (Gee, 2007). Serious games, in which the primary purpose is education or training, use entertainment value to entice players into learning about a wide variety of topics (Den Haan & Van der Voort, 2018; Lin et al., 2022). Players can take what they learn and immediately put it to use solving problems in the real world (Jenkins et al., 2006).

Games with a strong narrative component have the added potential to change individual real-world beliefs and attitudes (Gesser-Edelsburg & Singhal, 2013; Green, 2004). Transportation theory, which can be applied in a wide variety of media, describes how people can get lost in a narrative world (Green, 2004) and connect with characters in a story, thereby helping people better understand their own lives (Giles, 2002). Narratives are spaces for players to explore and take risks; the player, in the end, is safe, even though their game avatar is not (Green, 2004). Games “situate meaning in worlds of experience...that is ultimately shared, collaborative, social and cultural” (Gee, 2010, p. 189).

Games can also be social experiences (Gee, 2007) where multiplayer and collaborative games offer opportunities for learning (Den Haan & Van der Voort, 2018; Wang & Huang, 2021). The debrief, the discussion that occurs after the game, facilitates peer learning, with studies showing it is the point at which most learning occurs (Lederman, 1992; Schwägele et al., 2021). A structured, post-game debrief has two main purposes: (1) it gives players an opportunity to reflect on the game experience, make real-life connections, and learn from it

(Crookall, 2010, 2023; Lederman, 1992); and (2) it helps the facilitator (e.g., librarian, teacher) understand if the participants' takeaways match the learning objectives of the activity (Crookall, 2010; Lederman, 1992). Debriefing is not an “add-on” to the education experience but an essential component (Crookall, 2010, 2023; Lederman, 1992).

2.4 | Games and misinformation

Misinformation games have proliferated in recent years and cover a wide range of genres, have varied learning objectives, and exhibit different theoretical approaches (Contreras-Espinosa & Eguia-Gomez, 2023; DeJong, 2023a; Glas et al., 2023; Grace & Liang, 2023). Two review articles, Contreras-Espinosa and Eguia-Gomez (2023) and Glas et al. (2023), situate their analyses within media literacy frameworks—the Get Your Facts Straight! Toolkit and the Dutch Media Literacy Model 2021, respectively. While the authors do not refer explicitly to the sociocultural approach, the two frameworks contain elements that could be interpreted as such. For instance, the Dutch model emphasizes “that media literacy should not be seen as having one fixed outcome but rather should lead to the result of mutual and societal dialogue” (Glas et al., 2023). This disinclination towards a fixed outcome (e.g., discerning between real and fake content) is strongly reflected in the sociocultural approach. The Get Your Facts Straight! model has a module, titled “how social media make money and why misinformation and propaganda are vastly present on social media” (Contreras-Espinosa & Eguia-Gomez, 2023). This focus on infrastructural meaning-making is a core characteristic of the sociocultural approach.

Games based on inoculation theory deserve mention as a category that has received significant attention in both the popular press and academic literature. Inoculation theory approaches misinformation much like a medical vaccine—exposing individuals to weakened versions of arguments to build up resistance to future exposure (Roozenbeek et al., 2023). Bad News and Breaking Harmony Square are noteworthy examples. Studies of their effectiveness conform to the autonomous tradition of MIL with a focus on clearly identifiable and measurable learning outcomes, primarily related to misinformation detection (Basol et al., 2020; Modirrousta-Galian & Higham, 2023; Pimmer et al., 2020; Roozenbeek & van der Linden, 2020; Yang et al., 2021).

One observation we believe is consequential is that nearly all misinformation games are single player. In DeJong's landscape study of 32 digital misinformation games, all but one are single player, a finding that we

only ascertained after communicating with DeJong as this detail was not included in the original paper (personal communication, October 27, 2023). Considering the broad appeal of games as social experiences, and the fact that misinformation is largely a social media fueled phenomenon, this can be considered a missed opportunity for the field (Coward, 2023).

3 | THE GAME: THE EUPHORIGEN INVESTIGATION

The development of a misinformation escape room game—*The Euphorigen Investigation*—was motivated foremost by our earlier research on librarians' professional experiences with misinformation (Young et al., 2020). This study revealed, among other findings, a desire for interventions that addressed the affective dimensions of misinformation, were respectful of people's beliefs, could reach people who are most vulnerable to misinformation, approached misinformation indirectly, and encouraged dialogue among participants (Young et al., 2020). The idea for a misinformation escape room largely emerged from these findings.

Escape rooms are interactive, team-based games in which a group of players work together to solve puzzles and other challenges to achieve a particular goal in a limited amount of time (Makri et al., 2021; Veldkamp et al., 2020). Educational escape rooms have been used in formal and non-formal educational settings including public libraries (Makri et al., 2021; Veldkamp et al., 2020). We deemed an escape room to be a potentially excellent medium for learning about misinformation since the longer format of the game (typically 1 hour) would allow for narrative development, greater immersion in a fictional world of misinformation, collaborative problem-solving, a sense of achievement, and peer reflection and learning. Critically, we avoided an explicit skills-building orientation and instead opted to design an experience that revolved around a narrative and the affective dimensions of misinformation. One important consideration was how the game would work in a public library setting where participation in educational and other programs is voluntary, in contrast to a classroom environment where participation can be mandatory. Accordingly, it was important that our game would be enjoyable and appealing to a wide range of people in diverse communities. This criteria reinforced our decision to avoid skills-building which would likely be obvious to the participant and a detractor from the game's enjoyment. Our game designer collaborator, one of the founders of the escape room movement in the United States, played a critical role in this. Escape rooms

are primarily commercial enterprises, and thus similarly need to be enjoyable to attract customers.

In *The Euphorigen Investigation*, players must investigate a company's claims about its brain-boosting supplement, *Euphorigen*, to determine whether it is safe for the government to add it to the public water supply. Players solve a series of puzzles involving misleading headlines, misleading/manipulated charts, and deepfake images enroute to the goal of locating the video of a prominent scientist who supposedly has the definitive evidence. In the game's climax, players share the video with the public, believing they have completed the mission, only to discover that the video was a deepfake and that they must undo the damage.

Throughout the game, the game host provides encouragement, gives hints if a team gets stuck, and keeps time to ensure that each team completes the game in about 45 min to allow for a 15-min post-game debrief. The gameplay time is slightly shorter than conventional practice, but was requested by librarians to better fit their programming needs. In the debrief session, the game host leads players through a set of questions designed to elicit conversation and reflection. The debrief questions included, among others:

- This game involved misleading headlines, misleading/manipulated charts, social media bots, deepfake images, and deepfake videos. Was there anything new or surprising to you about these tactics?
- How did you feel when you learned that the video was a deepfake?
- You only had the option to share the videos. How much choice do you feel you have in real life on social media?
- In the game's final puzzle you were able to correct the mistake by sharing the real video. How do you think this would have played out in real life?

As mentioned above, players encounter several forms of misinformation throughout the game, including misleading headlines and charts, and deepfake images and videos. We specifically designed the game to be challenging and to lead players to believe misinformation as a way of eliciting feelings of vulnerability. This is especially prominent in the sequence where players inadvertently share a deepfake video, thinking they had solved the game. This design requirement was critical for addressing "third person effect" (the belief that one is less susceptible to misinformation than others). Players may feel embarrassed or regrettable after learning that it is a deepfake video, but this game action connects to an important debrief question about their feelings of sharing the deepfake (see above). The game host plays an active role in

the conversation by reiterating that everyone is vulnerable to misinformation, a fundamental concept in the misinformation literature, and suggesting that everyone has a role in supporting friends, family and others to resist misinformation. As such, the goal is to turn this experience into an empowering discussion.

4 | METHODS

As in the exploratory study (Cho et al., 2023), we chose public libraries as our research sites. Libraries have expressed interest in offering educational programming around misinformation (Alvarez, 2016), with librarians calling for more innovative approaches to misinformation education, recognizing that their existing efforts are inadequate (Paris et al., 2022; Tripodi et al., 2023; Young et al., 2020). Libraries also have a history of offering escape rooms for recreation and learning (Kroski, 2019). We employed a mixed methods approach, with a primarily qualitative orientation. This facilitates an analysis using common qualitative methods while also allowing for interpretation of the findings using the sociocultural lens. Librarian data came from game session reports and focus group discussions.

4.1 | Data collection

Working with the library training provider WebJunction, in late 2021 we advertised the game and research project in their newsletter. A total of 75 libraries applied for the opportunity to participate in the study by offering the game to patrons in their libraries. The final sample of 10 libraries were selected to achieve diversity across geographic, demographic, and political inclination. We excluded libraries that could not demonstrate sufficient readiness to host a virtual game. Since the previous study (Cho et al., 2023) focused on larger urban and suburban libraries, we purposefully focused on recruiting libraries from smaller and more rural locations.

For our study, the librarians performed multiple roles. First, they recruited players using their existing communications channels (e.g., newsletters). Second, the librarians served as the game host and debrief facilitator. Third, they were research subjects themselves. Due to the controversial nature of misinformation in some communities (Tripodi et al., 2023; Young et al., 2020), individual librarian participants were guaranteed confidentiality in publications.

We collected data from players and librarians using four methods approved by the University of Washington Institutional Review Board (IRB). Player data came from

two sources: debrief recordings and player survey responses. First, librarians recorded the debrief session in which they used a semi-structured debrief guide to facilitate an approximately 15-min discussion among the players immediately after completing the game. We requested librarians to adhere as closely as possible to the questions in the debrief guide for data analysis purposes. As previously discussed, the debrief included questions related to players' experiences with misinformation, their familiarity with the types of misinformation they encountered in the game, their reactions to sharing the game's deepfake video, and the real-life consequences of sharing misinformation. Second, the librarians directed the players to complete a 10-min online survey after the debrief. The survey covered such topics as: awareness of how misinformation spreads, attitudes towards misinformation, helpfulness of the game, and potential changes to future social media use. We also collected basic demographic information, social media usage, and ideological identification. Most questions were closed-ended with Likert scales; two open-ended questions allowed participants to provide comments.

Librarian data came from game session reports and focus group discussions. Librarians submitted a session report after each game using an online form, allowing them to report data and their perspectives about each game session, such as: completion time, descriptions of team members, technical issues, quality of debrief discussion, and any other insights or suggestions. Lastly, we held three focus group sessions for librarians to capture their overall experiences with hosting the game, their perceptions of how well the game achieved the learning goals, successes and challenges with the logistics of hosting the game, and their thoughts about fit for library programming on misinformation.

The research team held three training sessions for librarians in January 2022 that covered logistics and IRB procedures. Each librarian was approved by UW IRB to collect data from players on behalf of the research team. Each library hosted the game 4–7 times from February to April 2022. Libraries were tasked with scheduling the games and recruiting players. The 10 libraries hosted 53 sessions reaching 211 players (see Table 1). On average, the game session took 49 min to play with a range of 25–90 min, and the average session had four players. Players who completed the survey could enter a drawing for one of five \$25 gift cards. No personally identifying information was collected from the survey. The contact information needed for the gift card drawing was collected separately and not connected to that individual's survey response. The paper authors facilitated three focus groups with the participating librarians during late March and early April 2022. This concluded their

participation in the project. Nine libraries were provided \$500 as compensation for their participation in the study; one library declined payment.

4.2 | Data analysis

The authors led the data analysis, which included supervising student research assistants. The debriefs and player surveys were our main data sources, with the librarian focus groups and session reports providing further context. Audio and video recordings of the debriefs and focus groups were transcribed using a combination of human and machine-based methods. To assist with coding, the authors worked with a group of five student coders. We used a combination of inductive and deductive coding approaches using web-based ATLAS.ti software. An inductive/deductive hybrid approach to thematic analysis recognizes the inherent blurring between induction and deduction and harnesses the benefit of each through an “an iterative and reflexive process” (Fereday & Muir-Cochrane, 2006; Proudfoot, 2022). Thematic analysis in its separate inductive and deductive forms continues to be a commonly employed approach across multiple disciplines (Nowell et al., 2017). In our work, we chose to adopt the combined use of inductive and deductive approaches, as this hybrid approach to thematic analysis can help reinforce greater rigor to our analysis, as demonstrated by Fereday and Muir-Cochrane (2006).

The first author (Wedlake) and a student coder developed the first draft of the codebook, a modification of the one used for the exploratory study (Cho et al., 2023), based on our research questions and learning goals. Wedlake and five student coders coded the same debrief and used a consensus model in which coders compared and discussed their results to reach a consensus (Hill et al., 1997). After making adjustments, Wedlake and students independently checked the new codebook with a different debrief and added new themes and clarified code definitions. Wedlake and student coders coded sample sessions using paired coders for a final test of the codebook. After finalizing the codebook, under the supervision of the authors, the student coders coded independently and wrote analytical memos describing their reflections. “Awareness,” “reflection,” and “play” are example codes (see Appendix for full codebook). Author Wedlake then read all of the coders' analytic memos, reviewed the distribution of codes through the data set, and identified key themes within and across the codes that responded to our research questions (Braun & Clarke, 2006). Working with co-authors, themes were further developed and refined. The prevalence of these themes was then measured by the frequency of their

TABLE 1 Library game hosts.

Library name	State	Census location description	Number of game sessions	Total number of participants
Apache Junction Public Library	AZ	Rural, Fringe	5	13
Bartlesville Public Library	OK	Town, Distant	4	20
Bigelow Free Public Library	MA	Suburban, Large	4	16
Calcasieu Parish Public Library	LA	City, Small	5	16
Des Moines Public Library	IA	City, Small	6	28
Fresno County Public Library	CA	City, Large	6	28
Gwinnett County Public Library	GA	Suburban, Large	7	19
Jefferson County Library System (Boulder Branch)	MT	Rural, Remote	6	24
Jefferson-Madison Regional Library	VA	City, Small	6	32
William Jeanes Memorial Library	PA	Suburban, Large	4	15

appearance in the debrief sessions. Wedlake analyzed the quantitative data from the survey and session reports using descriptive statistics.

4.3 | Limitations

We opted to run the study as a field study in public libraries to understand how the game would perform in a realistic environment in public libraries. This decision necessitated that we give up some control over the administration of the study. Librarians recruited participants using their standard program marketing systems. Although we advised them to reach diverse populations, neither we nor the librarians had full control over the demographic makeup of participants. Moreover, while we selected libraries in racially diverse areas, we could not control who signed up for the game. Accordingly, we were unable to pursue an experimental design such as a random control trial. Additionally, there is inherently some variance in the experience across players. While we provided training for the librarians, natural variation in terms of personality, relationship with the players, facilitation skills, among other factors meant that the game experience was not uniform. This stands in contrast to the majority of misinformation game studies (based on inoculation theory) where there is no live game host or post-game discussion. Another limitation was our use of self-reported measures in the survey, such as their perceptions of their learning outcomes and future social media behaviors. Furthermore, there was no delayed follow-up to assess longer-term effects. Deploying the game in other more stable environments like classrooms may help collect more longitudinal data in future studies. Lastly, this study is only situated in public libraries in the

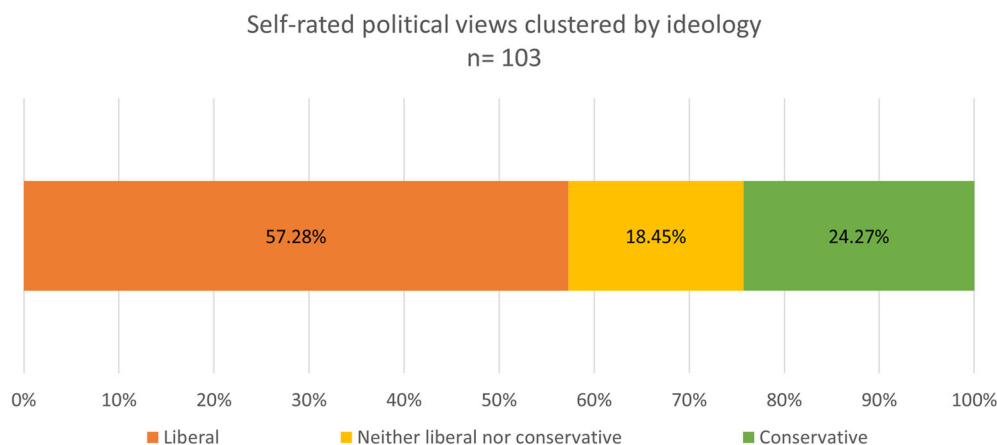
United States. As the game narrative was constructed to align with certain patterns of trust in institutions and historical parallels in the United States, we hypothesize that the game would function differently in dissimilar contexts. Recently, the game has been localized for other countries and in other languages. In addition to opportunities for future research on the game's effects, this also creates a need to develop flexible co-design methods that account for cultural and linguistic differences.

5 | PLAYER DEMOGRAPHIC SUMMARY

Out of 211 players, 127 completed surveys for a response rate of 60.19%. Respondents could skip any questions they did not want to answer. When calculating the percentages below, we removed the no responses from the calculations but note the denominator used. Respondents were majority female (61.76%, $n = 102$) and White (80.58%, $n = 103$). Respondents rated their political ideology on a seven-point scale from very liberal to very conservative. As shown in Figure 1, when clustered based on ideology, players skewed slightly liberal (57.28%, $n = 103$). The age range of the respondents was 14–82 ($n = 102$, mean = 39, standard deviation = 17). These data suggest the game was largely successful in reaching a broad range of participants across gender, age, and ideological categories. In 78% of game sessions, at least some of the players knew each other. In 40% of the sessions, everyone knew each other, and 38% of sessions had a mixture of players that knew and did not know each other.

Regarding social media use (Table 2), most respondents visit social media websites or apps at least once a

FIGURE 1 Self-rated political views clustered by ideology.



day (63.78%, $n = 127$). Survey respondents were less likely to visit news websites with only 29.13% ($n = 127$) visiting websites or applications at least once a day. Despite reported frequent social media use, the majority of respondents reported that they do not often share information online that they encounter on social media. Even fewer share information from online news sources (Table 2).

6 | FINDINGS

The insights from this study provide a perspective on how to use a sociocultural lens to understand the take-aways from a misinformation game intervention. Through playing the game, players (1) engaged in the process of infrastructural meaning-making to understand how affordances of social media platforms can spread misinformation; (2) reflected on the psychological influences that work in tandem with platform affordances; and (3) made connections to their role in the broader information environment. In each of these experiences, players worked collaboratively and shared their perspectives.

6.1 | Infrastructural meaning-making

During the debriefs, players expressed an understanding of how online technologies facilitate the intentional and unintentional spread of misinformation. The game's narrative and puzzles covered misleading headlines, misleading/manipulated charts, social media bots, deepfake images, and deepfake videos. A major discussion point during the debrief was the difficulty of determining whether AI-generated profile pictures were real or not. Despite the game providing guidance on techniques on how to identify these types of fake images, players

struggled to figure out which photos were real. A player from Pennsylvania Session 1 summed up the feelings of many participants, "I was surprised how terrible I was at figuring out if those pictures were real. That was really hard, super hard." The act of trying to determine the veracity of information gave players an appreciation for the sophistication of the technologies. Many players did not know that AI had the capabilities to create such real looking images. Some players discussed how they had known about these technologies, but the game gave them a new understanding for how the technologies can be used for purposes of spreading misinformation.

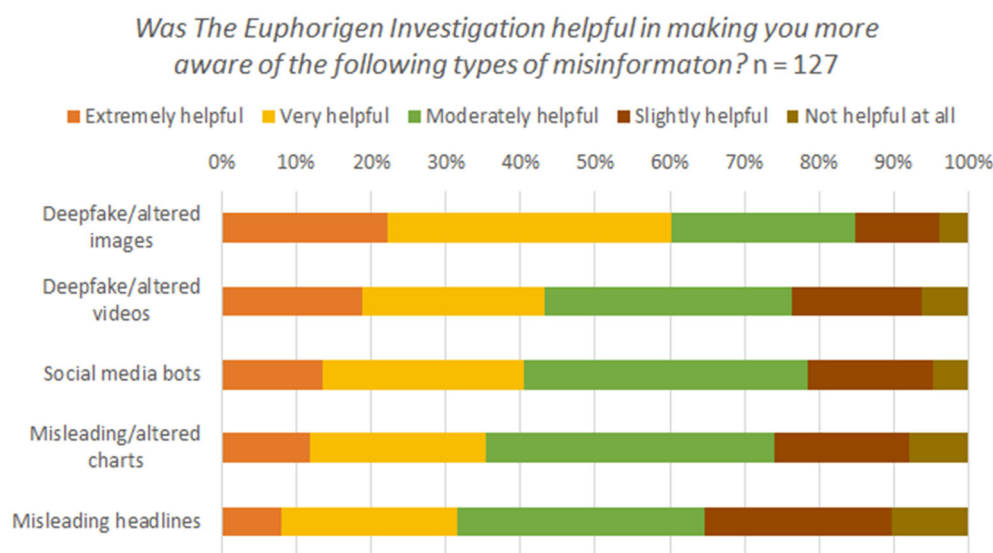
In the post-game survey, a majority of respondents said that *Euphorigen* was at least "moderately helpful" in making them more aware of particular types of misinformation (Figure 2). The ratings were highest for deepfake/ altered images (60.32% "extremely helpful" or "very helpful"), a not surprising finding since it is a newer technology. By the same token, the game was perceived less helpful for techniques that have been commonly used for a long time (e.g., misleading headlines, misleading/manipulated charts).

Players made connections from the fictional puzzles in the game to real-world events. For example, one participant in Iowa explained that these tactics have been highlighted in recent news topics:

You just had to look at the recent events in Ukraine and all the fake stuff that's being put out there about what the Russians are doing, what the Ukrainians are doing, what the various governments are doing and then a day later, they're back tracking and saying, oh, we had unreliable resources or that's fake or it's real. And I think with the advent of technology, this has been pretty commonplace for a few years now. (Speaker 2, Iowa Session 2)

TABLE 2 Social media and news consumption.

Self-reported frequency (<i>n</i> = 127)	Visiting social media	Online sharing from social media	Visiting news sites or apps	Online sharing from news sites
Several times a day	46.46% (59)	3.15% (4)	11.81% (15)	0.79% (1)
About once a day	17.32% (22)	6.30% (8)	17.32% (22)	3.15% (4)
A few times a week	13.39% (17)	17.32% (22)	29.13% (37)	7.87% (10)
Every few weeks	3.15% (4)	6.30% (8)	11.02% (14)	7.87% (10)
Less often	9.45% (12)	33.07% (42)	16.53% (21)	33.07% (42)
Never	10.24% (13)	33.86% (43)	14.17% (18)	47.24% (60)

FIGURE 2 Helpfulness of *The Euphorigen Investigation* in increasing awareness.

Although we did not have any follow-up surveys, during a librarian focus group, we received some evidence players might continue to make those connections after the game:

I do think that the wheels were turning because one of the sessions my husband was in it. And he said to me a few days later, “You know it’s really funny that I just played the game with you because they just had that fake [video] with President Zelensky.” He was thinking about it. (Librarian 1, Focus Group 1)

Setting the game in a fictional universe allowed players from different political and cultural perspectives to make meaning based on their own contexts. In addition to the Ukrainian/Russian war, during the debriefs, players made connections to a wide variety of topics such as fluoride in drinking water, COVID, and celebrities. In a librarian focus group, two librarians from a conservative area addressed why approaching the issues of

misinformation from a fictional perspective is particularly helpful in their community.

We’re also in the deep South and trying to overtly address misinformation is a little fraught...Something like this, opening that door and then following through with other resources is excellent, for our demographic for sure. (Librarian 3, Focus Group 1)

In another focus group,

The way it was presented took some of the political elements out so there was less disagreement or tension surrounding that. (Librarian 2, Focus Group 2)

One element in the game that differs from the real world is that the ending where players share the authentic video with everyone who received the deepfake. That is, they were able to simply undo the misinformation that had spread online. This was an intentional design

decision to ensure players have a sense of accomplishment and overall positive play experience. In the debrief, players consistently noted that this scenario would not play out the same way in real life. Players used terms like “spreads like wildfire” to describe how misinformation can spread online. In fact, five different players across five different debriefs used that exact phrase (Speaker 2, Massachusetts Session 4; Speaker 2, Massachusetts Session 5; Speaker 5, Montana Session 2; Speaker 4, Oklahoma Session 2; Speaker 3, Virginia Session 2). Responding to the survey prompt asking about their biggest lesson from the game, a player said,

The biggest lesson I learned is how impossible it can be to stop the spread of misinformation. Because of the [game's] smaller scale, I was really able to gain a perspective of how far a single video can reach across social media platforms or through social groups. (Respondent 7, California Session 3)

The game's fictional environment provided players with a safe space to understand how technologies and platforms operate in real-world contexts.

6.2 | Generating reflection on the psychological and emotional aspects of misinformation

In 21 of the debriefs, players described themselves as untrusting or cynical, and expressed general feelings of distrust around online information. Despite these players mentioning their distrust or skepticism at the beginning of the debrief, almost all of the players fell for the deepfake video and expressed feelings of vulnerability afterward. In this exchange from a debrief with a Californian librarian, the players describe some of the emotions they had when they found they had shared a deepfake video during the gameplay:

Librarian: Okay, when you learned that the video was a deepfake, that you guys have been “tricked.” How'd you guys feel?

Speaker 2: I was shocked.

Speaker 3: Honestly yeah, I wasn't expecting that. So, I was just like, Oh, dang.

Speaker 4: Yeah.

Speaker 2: We had just spent time looking at bots and stuff like that. So it's like, how did I not notice? Maybe it was just a really good deepfake... I don't know, I wasn't looking that close.

Speaker 3: I don't think I was paying attention... [overlapping conversation]. (California Session 2)

Having this shared experience of collectively falling for misinformation allowed the librarian host to open the conversation about everyone's experience with falling for misinformation in the “real world.” Some could recall specific situations but many, like this player, recognized that they probably did so without knowing:

Librarian: Have you ever believed something that was shared on social media that turned out to be untrue? Why do you think you fell for it? And how did it make you feel?

Speaker 4: I probably have at times, just 'cause a lot of times if I'm just going through [posts] really quickly and I see a bunch of people are sharing the same post, sometimes I'll hop on the bandwagon but honestly, it's not that I'm intentionally doing it, if it happens, it's because I'm literally scrolling through super fast, not even stopping to fully read. (Montana Session 2)

In both of these situations—gameplay and real life—players described how social media platforms create an environment that encourages users to passively and quickly consume information and not critically evaluate shared information.

The debrief also gave players an opportunity to reflect on the affective experience of misinformation. The players in an earlier quote described themselves as “shocked” or “tricked” after finding out they shared a deepfake video during the gameplay, and other affective descriptors included “betrayed,” “frustrated,” “surprised,” and so forth. The setup of the narrative (i.e., government/corporation deciding to inject a supplement in a public water stream which can make players feel that their personal choice is being taken away) is also deliberate in that we wanted players to have an emotional reaction that would impact how they process information encountered during the game. Some players made connections to how emotions can make them more susceptible to misinformation in the real world:

Speaker 4: There was a documentary a couple of years ago that Animal Planet did that was like, “Oh, we found mermaids.” and I’m not gonna lie y’all at first I was like, “Yes, they finally found them.” I guess something that makes life seem more or less bland than it really is, I tend to get hooked really bad on that kind of stuff.

Speaker 3: Isn’t it also when we get emotionally involved with something that we’re more likely, if it pulls on our heart strings, to believe it?

Speaker 4: Yeah, I would imagine. So the stronger your emotions, the stronger you’re gonna connect to some things. (Louisiana Session 3)

Speaker 4 from Louisiana, Session 3 described being more susceptible to getting “hooked” on information when it triggers stronger emotions in people. Speaker 3 brought in some prior knowledge to make the connection to emotionally charged content to belief.

The quantitative data lend further support to these findings. From the survey, the game increased players’ feelings of vulnerability and susceptibility to misinformation (Figure 3). After the game, a majority of players said they were more concerned about misinformation, and almost half of the players were more worried about believing (48.81%) or sharing (49.42%) misinformation. 73.23% of players were more concerned about misinformation in society after playing the game.

One interesting finding, found in the qualitative and quantitative data, concerns people’s confidence in identifying misinformation. One goal of the game is for players to reflect on the impracticality of relying on simple detection skills when it comes to sophisticated AI generated content. Given this, we see a tension or contradiction in the data between people who “feel more confident in my ability to identify misinformation” (70.86% strongly or somewhat agree; Figure 3) along with statements in the debriefs that similarly express greater confidence, and data (qualitative and quantitative) demonstrating the opposite: that detection was difficult for such technologies. This could indicate that people may have felt more confident in detecting certain types of misinformation such as misleading information in the manipulated headlines and charts puzzles, and less confident regarding deepfake images and videos.

While many players expressed greater appreciation for the sophistication of the technologies, a subset of players interpreted the goal of the game is to make people suspicious and critical of social media.

Speaker 2: So is the point of this thing to make people more suspicious of media, ‘cause that’s kind of what it’s trying to do.

Librarian: I think the point is not to make you suspicious so much, as make you stop and think or count to ten before you press the share button.

Speaker 2: Before you use share button. No. I don’t do a lot of Facebook stuff, so I don’t share a lot of stuff.

Librarian: Right. There’s so many ways to share, even recently there was something through the library network about our two of our delivery guys and it was crazy. (Massachusetts Session 2)

This librarian was able to redirect the conversation back to the core message, demonstrating the importance of a knowledgeable and trusted game host, but it is equally the case that some players may have come away from the game feeling less confident or more resigned to the power of these technologies and the ways social media facilitates the flow of misinformation. We return to this issue of individual agency in the discussion section below.

6.3 | Connecting personal behaviors to the global information environment

Many players understood they have an individual role in stopping the spread of misinformation. They realized the real possibility that they had inadvertently shared misinformation in the real world. The post-debrief survey asked participants to recount the biggest lesson from the game. This Arizona player’s survey comment exemplifies an ideal outcome, something we found frequently in the data:

Even though I don’t have a huge scope of influence online, what I post matters. I have a duty to look into the claims of things I may report, because I do not want to contribute to the spread of misinformation. (Respondent 8, Arizona Session 3)

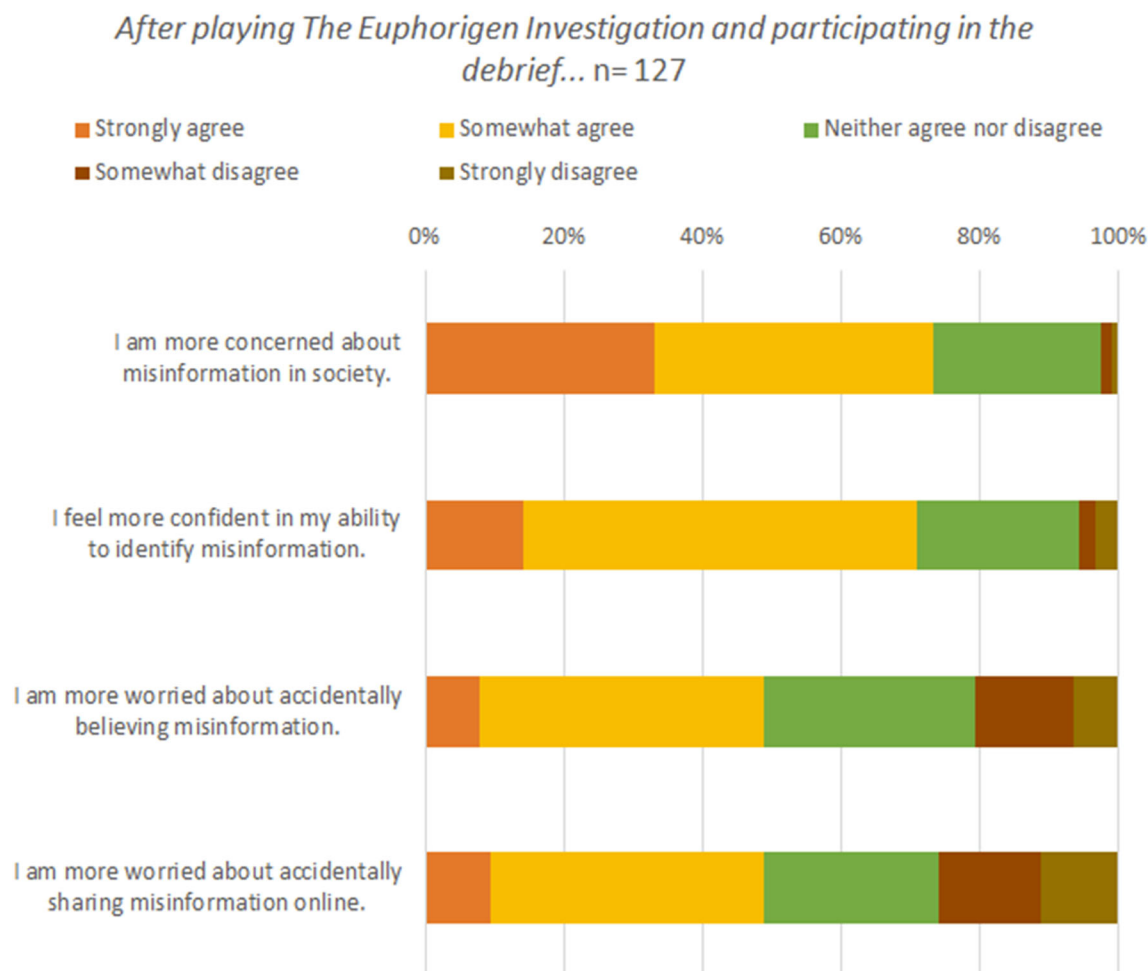


FIGURE 3 Players' perception after playing *The Euphorigen Investigation* and participating in the debrief.

Other players felt that by understanding the ease at which people can believe misinformation and how quickly misinformation can spread, the game has given them “a lot to think about regarding my sharing activities online” (Speaker 4, Georgia Session 6). At the same time, a few players also noted the difficulty of putting some of the lessons into practice in their everyday lives:

I think it's very easy to go ahead and just share something that you're inclined to want to believe. We all have our biases and pre-conceived notions, and we jump on that bandwagon, and so I think this [game] is really good to bring more awareness to taking the time on this material that can really influence people's mindset to take a little time and really do your study. But of course, we live in the modern era, we're all trying to

run with the rat race and doing the research, at the time, that can be an issue for a lot of folks. So I can see how misinformation can be so easily spread via social media, and just the fact that we don't have time to do our own sleuthing. (Speaker 5, Virginia Session 4)

The above speaker described how psychological biases interplay with how we consume information, then connected this to the difficulty of navigating the online ecosystem.

However, the debrief and survey also revealed a notable portion of players who believe their current online behaviors insulate them from engaging with misinformation—either through careful use or not engaging in political activities. The post-debrief survey asked, “After playing *The Euphorigen Investigation*, do you think you will change the way you use social media?” After removing the “never” responses regarding

the use of social media, the overwhelming response was “no” and with “yes” and “unsure” almost split evenly (Figure 4).

To further unpack this question, the survey offers an open-ended question for respondents to elaborate. After categorizing the open ended responses paired with the “no” response, the following explanations stood out for not changing their social media behavior (Figure 5): “already aware and/or cautious using social media” (34.78%), “do not use social media much” (23.19%), “do not think it applies to me because, for instance, only use it to keep up with family and friends” (21.74%).

In the debriefs, some participants discussed how they felt their social media use did not make them vulnerable to misinformation, as in this exchange.

Speaker 1: Yeah, when I joined Facebook, I thought it was mainly family, friends, and Twitter, I don't do Twitter. I thought that was always more political, so I've kind of avoided Twitter.

Librarian 1: Yeah, it can be really political, but it is surprising how much you see even on Facebook and Instagram.

After playing *Euphorigen*, do you think you will change the way you use social media?
(never SM users removed) n = 113

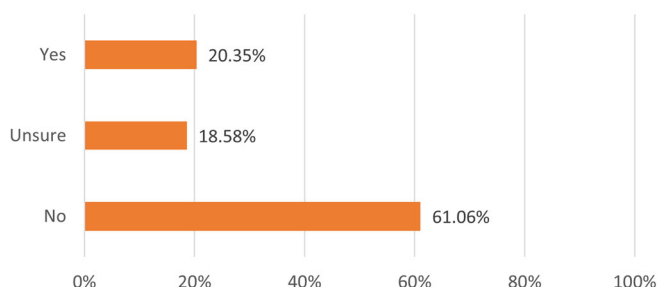


FIGURE 4 Anticipated change in social media use.

Categorized reasons why players will not change their social media use, n = 58

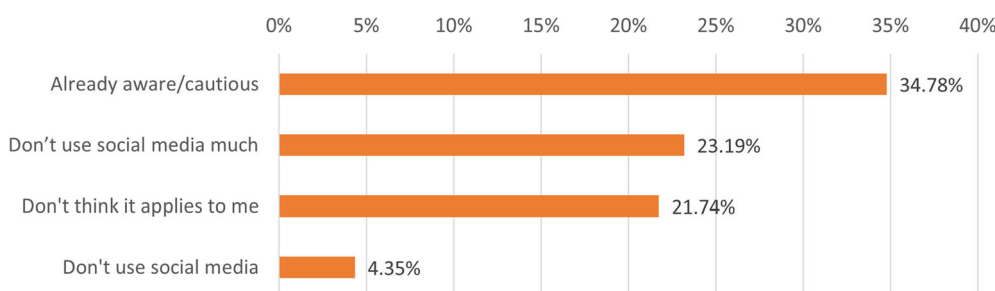


FIGURE 5 Reasons players will not change the way they use social media.

Speaker 1: One of my daughters is very political on Facebook, and so I kind of just skip over those, but I... Sometimes, she posts real things about my grandkids, I'm all over that. (Arizona Session 1)

This player made tactical decisions to avoid misinformation, such as not using Twitter and skipping over political posts from her daughter. Other participants said they only use social media for “fun stuff” like “cat videos,” so they do not personally worry about misinformation. Others reported that they do not share on social media. These statements align with the substantial percentage of survey respondents who reported that they “never” share on social media (33.86%) or via online news sites (47.24%). This points to the importance of helping people better understand the information environment, such as how their presence and engagement on social media, even without liking or sharing content, influences the information they and others see. Librarians believed narratives with less emphasis on news would help address this shortcoming. Suggestions from librarians included more of a focus on how people use apps like Instagram or interactive games like Roblox. One librarian (Focus Group 3) shared that in their library programs focused on the “paranormal or ghosts” are very popular at their library and “you'll get as twice as many that signed up show up.”

7 | DISCUSSION

The nationwide study of *The Euphorigen Investigation* provides an in-depth understanding of how the game supports learning about misinformation, our first research question. We found that the game helps increase awareness of misinformation technologies and encourages reflection on the affective dimensions of misinformation and the role of misinformation in people's lives.

7.1 | How The Euphorigen Investigation supports misinformation education

We found that players gained greater understanding of the ways platform infrastructures facilitate the spread of misinformation. Our study extends prior work (Cho et al., 2023) by centering infrastructural meaning-making (Haider & Sundin, 2022) as a primary learning outcome for a misinformation game. After the game, players exhibited a greater awareness of the sophistication of AI-generated content and the impracticality of always being able to discern true from false information on one's own. Players also made connections to their own online behaviors—past and present—to the spread of misinformation. These everyday experiences of using information are a critical component of the sociocultural perspective.

Players also understood how social media platform infrastructures exploit psychological and emotional tendencies that encourage the belief and sharing of information by individuals. The game simulated the experience of “lazy thinking” promoted by social media platforms that make players more vulnerable to misinformation (Pennycook & Rand, 2019; Schwarz & Jalbert, 2021). After playing the game, some players expressed the intention to use social media more deliberately. The debriefs were rich in discussion about how players fell for misinformation in the game and in real-life because “they fail to think” (Pennycook & Rand, 2019, p. 48). Many open-ended survey responses stated their intention to change such as “be more careful” or “think twice.” On the other hand, some players described themselves as inherently distrustful, and in the debriefs said the tactics demonstrated in the game gave them even more reason to distrust online information. This tension between skepticism and cynicism as documented in previous research needs careful attention (Clayton et al., 2020; Modirrousta-Galian & Higham, 2023; Pinkleton et al., 2012). As shown earlier, librarians play an important role as the facilitator to mitigate against a fall into cynicism, and supporting materials should emphasize this role.

Regarding the second research question on considerations that can be derived from the sociocultural perspective to inform the design and study of misinformation MIL interventions, we turn to a discussion of specific aspects of the game: namely, the narrative, the debrief, and collaboration.

7.2 | Narratives in games

Our study builds upon transportation theory which posits that an immersive narrative can be effective in evoking

emotions and connecting to real-world learning (Gesser-Edelsburg & Singhal, 2013; Green, 2004). Using a game format to educate people about misinformation allowed players to have fun and learn through trial and error in a friendly, low risk setting (Green, 2004; Jenkins et al., 2006). The narrative evoked emotions and cognitive biases that people experience when using online platforms (Schwarz & Jalbert, 2021). We found that creating a realistic, yet fictional setting helped players connect with the narrative without directly engaging with divisive topics. The fictional setting further allowed the game to appeal to individuals across the ideological spectrum. As other research has shown (Tripodi et al., 2023; Young et al., 2020), public libraries are conscientious about serving people in their communities who hold different beliefs. Designers of games and other MIL interventions that employ narrative techniques should be strategic when creating the setting, characters, and plot so as to evoke the targeted reactions while also providing sufficient narrative distance for players to have fruitful conversations within a group of people who may or may not share common beliefs.

7.3 | Debrief

The debrief discussions allowed participants to process their experience of playing the game, thereby “turning it into learning” (Crookall, 2010, p. 907), and to share and build on their learning from others (Crookall, 2010, 2023). The debrief gave players space to think and reflect after the gameplay had finished. Librarians, those leading the debrief, played critical roles in the overall experience. Librarians as game hosts can bring their subject-matter expertise and knowledge of information literacy into the discussions, in turn allowing them to understand what players learned and why (Lederman, 1992). During the debrief, players made numerous connections to current events and to the social media and news platforms they engage with in everyday life, and reflected on the emotional reactions and psychological biases that misinformation can trigger. The debrief is as critical as the game itself, not an optional component.

7.4 | Collaboration

Misinformation is not a problem to be solved only by individuals discerning what information is accurate and simply taking care of themselves (Ma, 2021). Social media is participatory and requires the knowledge of how to interact with others as part of a broader community (Jenkins et al., 2006). The benefits of a collaborative game

were apparent throughout the study. Collaborative games offer distinct learning opportunities (Den Haan & Van der Voort, 2018; Wang & Huang, 2021). Working together in the game and in the debrief allowed players to share knowledge with one another, including making real world connections from the fictionalized game setting. The escape room format provides a shared experience with the debrief providing space for players to discuss the game, share personal stories and strategies, and reflect on their own social media behaviors. While this study did not involve a delayed follow-up, games have the potential to be memorable experiences which allow for future reflection and conversation. The anecdote from the librarian about her spouse continuing the discussion about the game a few days later suggests this is an area for further study.

8 | CONCLUSION

The Euphorigen Investigation shows promise of a collaborative and immersive game to support learning about misinformation. In particular, the sociocultural lens of media and information literacy offers different ways to understand the game's role in this learning by focusing on such dimensions as infrastructural meaning-making and context. Certainly, one game (or any educational approach or resource) will only have a limited impact on such a multifaceted and complex challenge as misinformation. There must be continued efforts to engage people in conversations around misinformation, and to keep pace with technological advances. For instance, since *The Euphorigen Investigation* was developed, more sophisticated AI tools such as ChatGPT have emerged.

While our game had broad appeal with a narrative everyone could relate to, librarians reported that more targeted narratives would enhance the game's appeal for certain communities. In response, the research team undertook a number of co-design projects that focus on misinformation tactics prevalent in various online communities, such as K-pop fandom, breast cancer patients, and cryptocurrency schemes that target marginalized populations. Future research will explore whether such contextual targeting has an impact on player learning outcomes.

Lastly, the sociocultural perspective offers an important addition to the overwhelming focus of misinformation MIL interventions based on the autonomous model of MIL, which views literacy as the individual acquisition of skills, irrespective of context. As described earlier, studies of misinformation games have almost exclusively focused on such individual measures. This study

demonstrates how a sociocultural perspective can reveal important insights that otherwise may be obscured by the narrower focus of the autonomous perspective.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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